

PES UNIVERSITY

Department of Computer Science and Engineering

UE24CS341A: Software Engineering

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Project: Personal Subscription & Recurring Expense Auditor

Name: ES Sukruth

SRN: PES1UG24CS160

Submission Date: 8 September 2026

1. Introduction & Objective

The objective of this lab was to take the requirements produced in Lab 1 for the Personal Subscription & Recurring Expense Auditor and turn them into a working Agile plan in Jira. Concretely, this meant building a product backlog of Epics and User Stories, prioritising those stories, estimating them with story points, organising the work into two sprints, and then simulating the execution of those sprints by moving work items across the board. Finally, the burndown charts were used to reflect on how the plan played out.

This report documents the Jira project that was built, the backlog and its prioritisation, the story-point estimates, the two-sprint simulation, the burndown analysis, and answers to the reflection questions set in the lab handout.

2. Jira Project Overview

The work was carried out in a company-managed Scrum project in Jira.

- Space / Product: Personal Subscription And Recurring Expense Auditor
- Board: PSREA1 board (Scrum)
- Work item key prefix: PSREA1
- Total Epics: 4
- Total User Stories: 9 (PSREA1-5 to PSREA1-13)
- Total Story Points in backlog: 38
- Sprints: 2 (Sprint 1 = 23 points, Sprint 2 = 15 points)

3. Product Backlog – Epics

The requirements were grouped into four Epics, each representing a major feature area of the product:

Epic	Description	Related Requirements
Epic 1: Transaction Import & Analysis	Importing the user's transaction data, parsing it securely, and detecting recurring subscription charges.	FR-001, NFR-001, NFR-002
Epic 2: Subscription Alerts & Notifications	Alerting the user before renewals and showing upcoming renewal dates.	FR-002
Epic 3: Cancellation & Subscription Management	Viewing subscription details and providing guidance/links to cancel unwanted subscriptions.	FR-004
Epic 4: Audit Reporting & Export	Producing and exporting audit reports and summaries of subscription spending.	FR-005

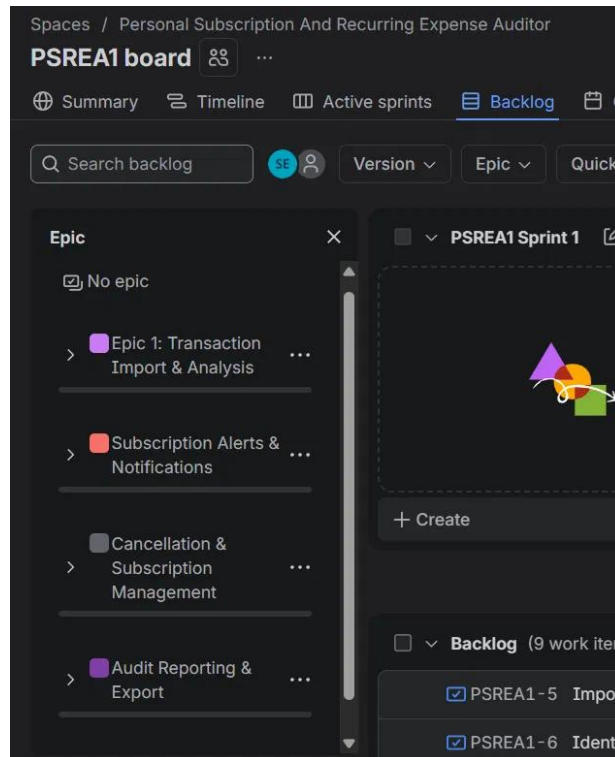


Figure 1: The four Epics created in the Jira backlog.

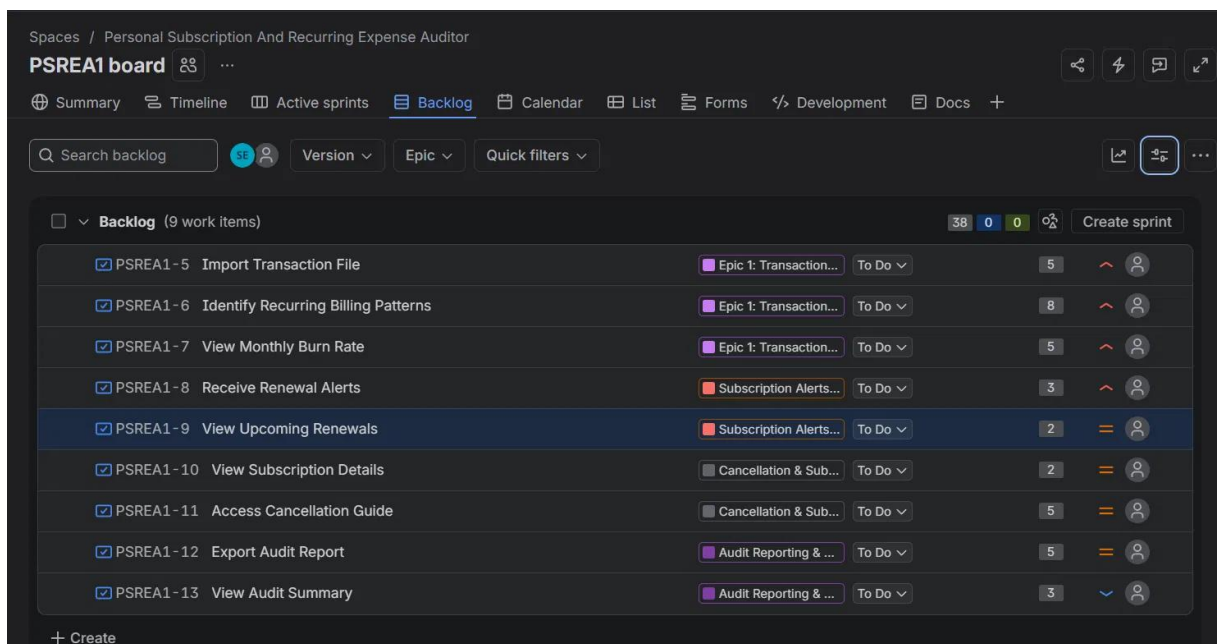


Figure 2: Product backlog showing all 9 user stories grouped under their Epics, with story points and priorities.

4. User Stories

Each Epic was broken down into user stories written in the standard “As a / I want / So that” format. The nine user stories are listed below.

ID	Story	Epic	Description (As a / I want / So that)
PSREA1-5	Import Transaction File	Epic 1	As an Individual User, I want to upload my bank/transaction file (CSV/OFX), so that the system can analyse my spending.
PSREA1-6	Identify Recurring Billing Patterns	Epic 1	As an Individual User, I want the system to automatically detect recurring subscription charges, so that I can see all my active subscriptions in one place.
PSREA1-7	View Monthly Burn Rate	Epic 1	As an Individual User, I want to see a monthly burn-rate summary grouped by service category, so that I can understand how much I am spending on subscriptions each month.
PSREA1-8	Receive Renewal Alerts	Epic 2	As an Individual User, I want to be notified at least 7 days before a subscription renews, so that I can decide whether to keep or cancel it.
PSREA1-9	View Upcoming Renewals	Epic 2	As an Individual User, I want to see a list of upcoming renewals, so that I can plan for forthcoming charges.
PSREA1-10	View Subscription Details	Epic 3	As an Individual User, I want to view detailed information about each subscription, so that I can understand its cost and billing cycle.
PSREA1-11	Access Cancellation Guide	Epic 3	As an Individual User, I want a step-by-step cancellation guide with a direct link, so that I can easily cancel unwanted

			subscriptions.
PSREA1-12	Export Audit Report	Epic 4	As a Finance Auditor, I want to export a full audit report, so that I can review the spending records offline.
PSREA1-13	View Audit Summary	Epic 4	As a Finance Auditor, I want to view a summarised audit of subscription spending, so that I can quickly assess financial exposure.

An example of a fully detailed story in Jira is shown below.

The screenshot displays the Jira interface for a story titled "View Monthly Burn Rate" under the project PSREA1-7. The main content area includes a description: "As an Individual User, I want to see a monthly burn-rate summary grouped by service category, So that I can understand how much I am spending on subscriptions each month." Below the description are sections for "Subtasks" (Add subtask), "Linked work items" (Add linked work item), and "Confluence content" (Product requirements). The right sidebar contains the "Details" section, which lists: Assignee (Unassigned), Reporter (SUKRUTH ES), Labels (Add labels), Due date (Add due date), Start date (Add date), Story Points (5), Priority (High), and Parent (PSREA1-1 Epic 1: Transactional). The top navigation bar shows the breadcrumb "Spaces / Personal Subscription... / PSREA1-1 / PSREA1-7" and the user "sukruthes27@gmail.com".

Figure 3: Story detail for PSREA1-7 "View Monthly Burn Rate" – description, story points (5) and priority (High).

5. Prioritisation & Story Point Estimation

Stories were prioritised (High / Medium / Low) and estimated on the Fibonacci scale (2, 3, 5, 8). High-priority, high-value core stories were placed at the top of the backlog; supporting and reporting stories were ranked lower.

ID	Story	Priority	Story Points
PSREA1-5	Import Transaction File	High	5
PSREA1-6	Identify Recurring Billing Patterns	High	8
PSREA1-7	View Monthly Burn Rate	High	5
PSREA1-8	Receive Renewal Alerts	High	3

PSREA1-9	View Upcoming Renewals	Medium	2
PSREA1-10	View Subscription Details	Medium	2
PSREA1-11	Access Cancellation Guide	Medium	5
PSREA1-12	Export Audit Report	Medium	5
PSREA1-13	View Audit Summary	Low	3
	TOTAL		38

The largest estimate (8 points) went to “Identify Recurring Billing Patterns” because detecting recurring charges across varied billing cycles is the most algorithmically complex piece of work. Simple lookup/display stories such as “View Upcoming Renewals” and “View Subscription Details” were the smallest at 2 points each.

6. Sprint Planning & Simulation

The 9 stories were divided across two sprints. Sprint 1 focused on the core value features (import, detection, burn-rate and renewal alerts); Sprint 2 covered the remaining management and reporting features.

6.1 Sprint 1 (23 story points)

Sprint 1 committed to 5 stories totalling 23 points: PSREA1-5 (5), PSREA1-6 (8), PSREA1-7 (5), PSREA1-8 (3) and PSREA1-9 (2).

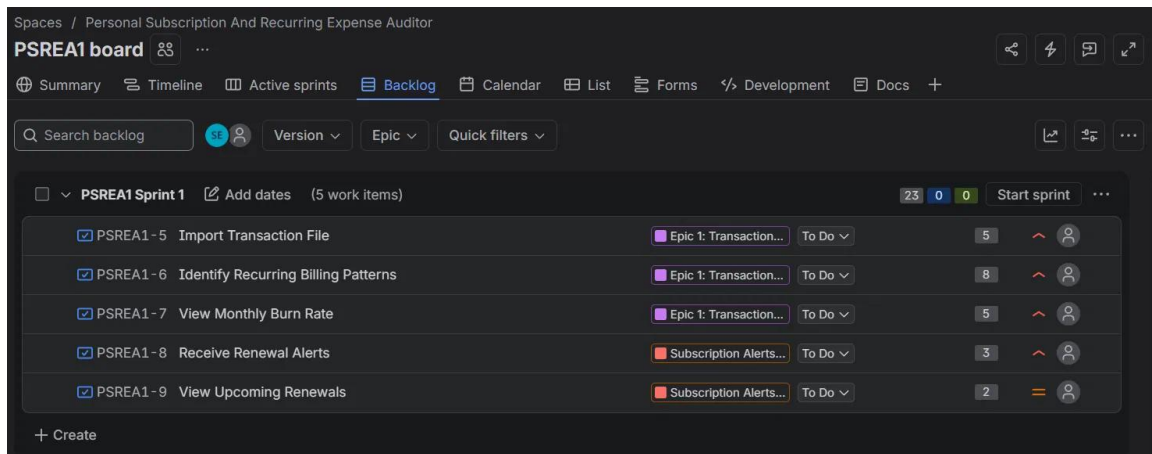


Figure 4: Sprint 1 planned with 5 work items (23 points) before starting the sprint.

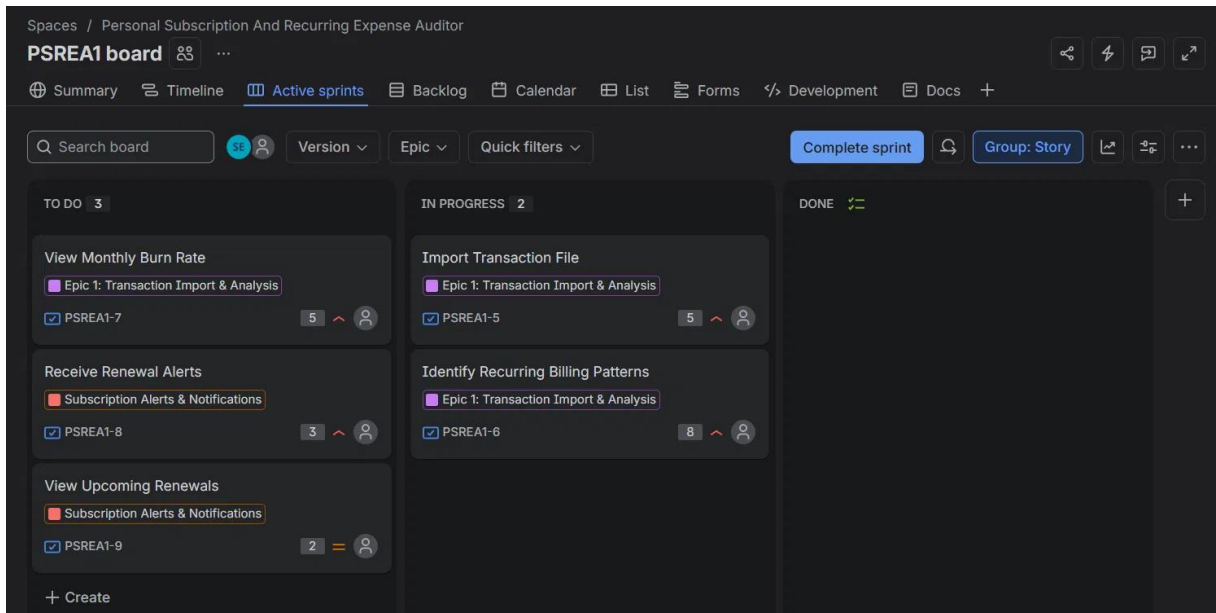


Figure 5: Sprint 1 board mid-execution – work items moving through To Do, In Progress and Done.

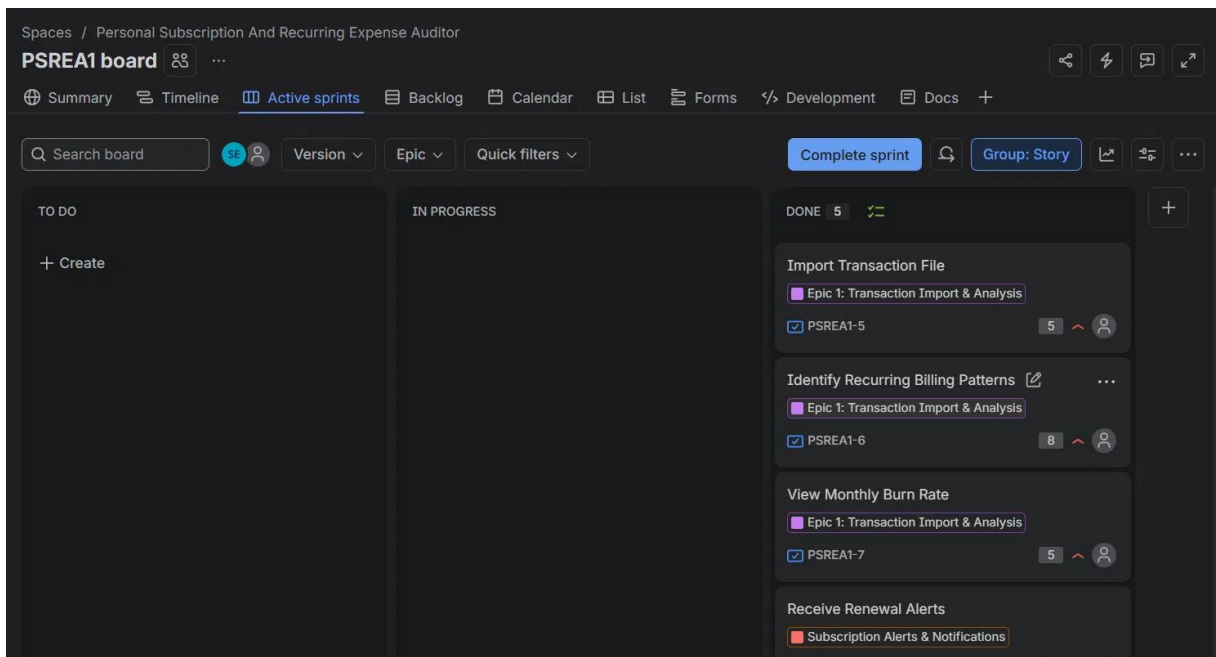


Figure 6: Sprint 1 board at completion – all 5 stories moved to Done.

6.2 Sprint 2 (15 story points)

Sprint 2 committed to the remaining 4 stories totalling 15 points: PSREA1-10 (2), PSREA1-11 (5), PSREA1-12 (5) and PSREA1-13 (3).

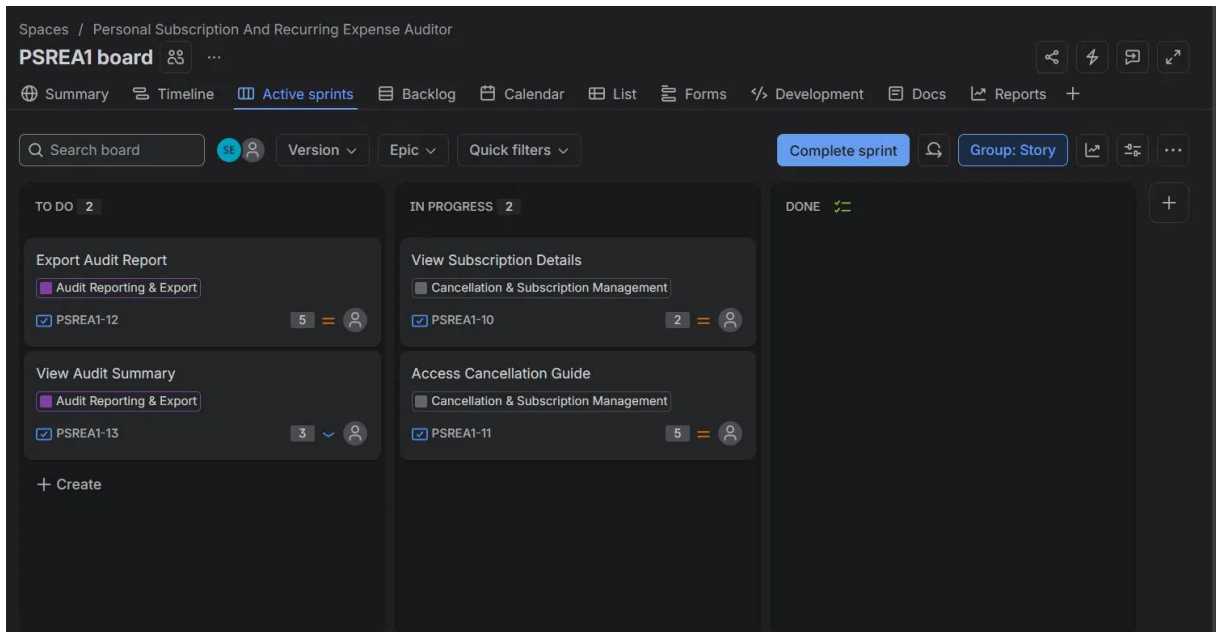


Figure 7: Sprint 2 board – stories progressing from To Do into In Progress.

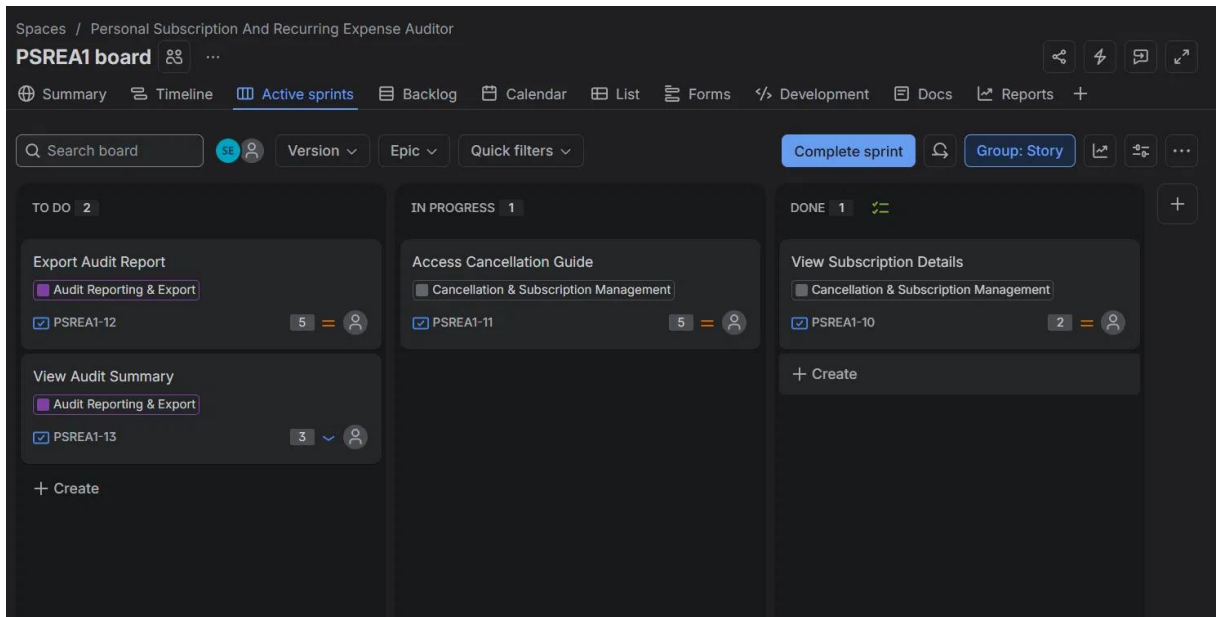


Figure 8: Sprint 2 board – first stories reaching Done as the sprint progresses.

7. Burndown Chart Analysis

A burndown chart was generated for each sprint. The grey line is the ideal guideline; the red line is the actual remaining story points.

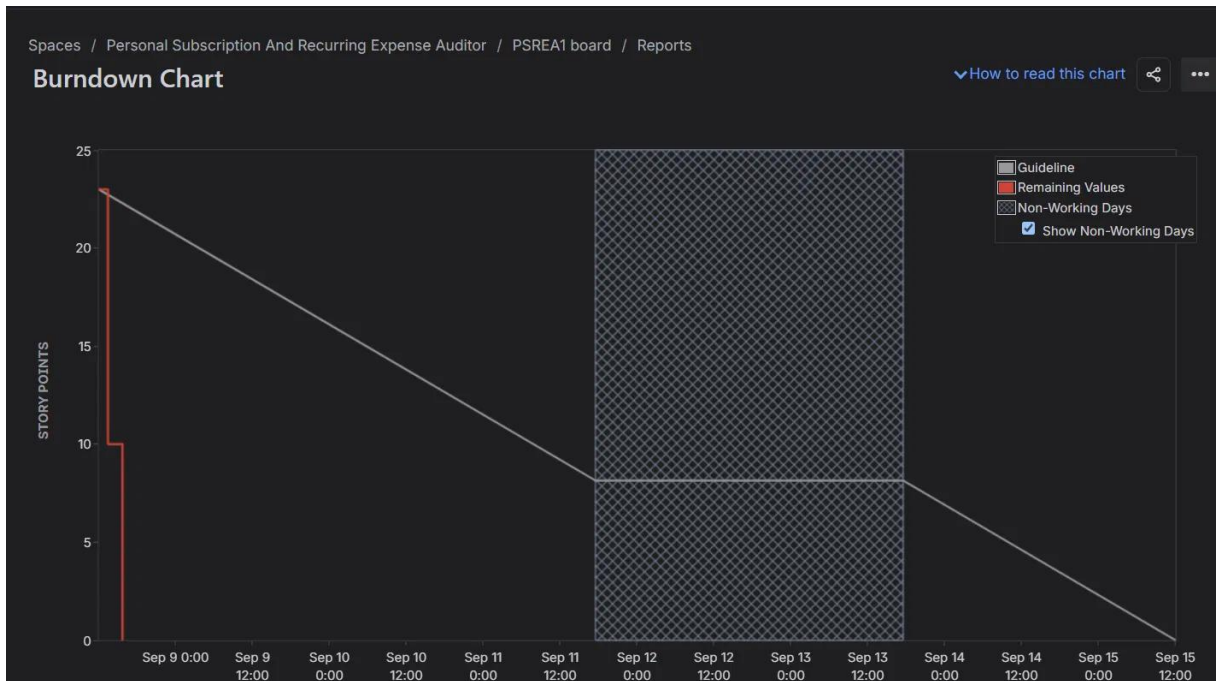


Figure 9: Sprint 1 burndown – remaining work drops from 23 points to 0.



Figure 10: Sprint 2 burndown – remaining work drops from 15 points to 0.

In both charts the red remaining-work line falls sharply to zero, well ahead of the grey guideline. This reflects the compressed, single-session nature of the simulation rather than a steady day-by-day burn, but it confirms that all committed work was completed in each sprint.

8. Reflection Questions & Answers

Q1. Did your story point estimates reflect the actual effort and complexity of the work?

Yes. The Fibonacci scale captured relative complexity well. “Identify Recurring Billing Patterns” (PSREA1-6) was rated highest at 8 points because pattern detection across varied billing cycles is the most complex piece of logic, while simple display/lookup stories such as “View Upcoming Renewals” and “View Subscription Details” were correctly the smallest at 2 points. During the simulation the higher-pointed stories did feel proportionally larger, so the estimates held up.

Q2. Was your product backlog well-prioritised?

Yes. High-priority, high-value core stories (transaction import, recurring-charge detection, monthly burn-rate and renewal alerts) were placed at the top of the backlog and pulled into Sprint 1, while Medium/Low supporting features (subscription details, cancellation guide, audit export and audit summary) were ranked lower and scheduled for Sprint 2. This front-loads the features that deliver the product’s primary value – knowing what you are spending and when things renew – before the convenience and reporting features.

Q3. How did the simulated sprint execution align with your sprint plan?

Very closely. Sprint 1 committed to 5 stories / 23 points and every story moved To Do → In Progress → Done with no carry-over. Sprint 2 committed to 4 stories / 15 points and was likewise completed. Moving the cards across the board mirrored the planned scope exactly, which shows that both sprints were realistically scoped for the committed capacity.

Q4. What insights did the burndown chart give you about capacity and progress?

Both burndown charts show the remaining-work line dropping sharply to zero ahead of the ideal guideline, which is expected given the compressed one-session simulation. More usefully, comparing the two sprints highlights an uneven workload – Sprint 1 carried 23 points versus Sprint 2’s 15. In a real project this suggests balancing the work more evenly across sprints to achieve a steadier, more predictable velocity. It also shows how the guideline-vs-actual comparison acts as an early-warning signal for scope or capacity problems.